

CCST9012 Tutorial Week 1

Use of gnomon for tracking the Sun

In Lecture 2, we discussed the use of an early technology for tracking the sun throughout a day over the course of a year called a **gnomon**.

Open the Gnomon geogebra simulation.

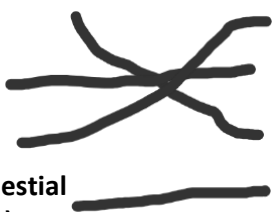
You should see two views. On the left, you see a view looking down at the ground from above the Gnomon.

On the right you see a horizon intersecting the celestial sphere. In the very center is a gnomon.

Below this view, the values of Altitude and Azimuth refer to the position of the sun in the sky as seen by an observer standing at the gnomon.

By default, a slider for *Latitude* should be set to 23 degrees. You may increase it to 38 degrees for Athens.

Q0. Match the term to its definition.

Terms		Definitions
Horizon		A line across the sky that runs from North to South.
Zenith		The point in the sky directly above the observer's position.
Meridian		A plane perpendicular to the gnomon.
North Celestial Pole (NCP)		The point in the sky around which the stars rotate.

We have learned that the path of the sun changes over the course of a year. These paths may be explored by shifting the position on the *Annual Motion* slider. On the right side of the simulation, the sun's daily motion is shown as an orange-colored circle.

Inclination refers to the angle the daily path of the sun makes with the horizon.

Q1. How does the inclination change throughout a year?

The inclination **does not change** throughout a year.

Shift the *Annual Motion* slider all the way to the left.

Q2. What is the *azimuth* at sunrise?

The azimuth is 122.44 degrees.

Click the box labeled *Show Trace*. On the right, you will see the sun follow its daily path. On the left, you will see red dots that trace the location of the gnomon's shadow.

Using the *Daily Motion* slider, position the sun until it is crossing the meridian.

Q3. In which direction does the shadow point when the shadow is the shortest?

The shadow points to the North.

Q4. What is the position (azimuth, altitude) of the sun?

The altitude of the sun is 28.6 degrees, while the azimuth is 180 degrees (due South).

Shift the *Annual Motion* slider all the way to the right.

Q5. What is the azimuth at sunrise?

The azimuth is 62.21 degrees. This is further north than when the annual motion slider is all the way to the left.

Q6. What is the altitude and azimuth of the sun at noon?

As before, the azimuth is 180 degrees (due South) while the altitude is much higher, 75.4 degrees.

Click the *Show Trace* box for a few more positions of the *Annual Motion* slider.

Q7. For what position of the annual motion slider do the red dots form a straight line?

The red dots (trip of the sun's shadow) form a straight line when the Annual Motion slider is in the very center.

When we click the Show Trace button, we animate the sun from shortly after sunrise to shortly before sunset, when the sun casts a well defined shadow on the gnomon. However, the rest of the sun's path is shown as a full circle.

Q8. Compare what fraction of the sun's path is above the horizon to below for different positions of the Annual Motion slider. For which position is the length of daylight the longest? Shortest?

When the Annual Motion slider is all the way to the right, more of the Sun's daily path is above the horizon while the opposite is true when the annual motion slider is all the way to the left.

Recall the *Tropic of Cancer* is a line of northern latitude that separates locations that have a *Zenith crossing* from those that do not.

Q9. Use the simulation to determine the latitude of the Tropic of Cancer.

We can shift the Latitude slider until we find the latitude where the point on the sun's daily path labeled Maximum Altitude coincides with the observers Zenith. This happens around 23 degrees of latitude.

Q10. How is the relationship between the inclination of the sun's daily paths dependent on latitude? Find the latitude where this angle is 90 degrees.

This angle is larger when the latitude is smaller. It is 90 degrees with the latitude is 0 degrees.

Set the latitude to 22.5 degrees, appropriate for Hong Kong.

Q11. Compare your answers to Questions 2-7 between Hong Kong and Athens.

2. 116.54 degrees. The sunrise is more south of east in Athens than in Hong Kong.

3. North, same as in Athens.
4. The azimuth is the same as in Athens, 180 degrees, but the altitude is higher, 44.1 degrees.
5. The azimuth of sunrise is 66.09. The sunrise is further north of east in Athens than in Hong Kong. We can conclude that the annual shifting of sunrise locations in Athens is more extreme than in Hong Kong.
6. The azimuth is 360 or 0 degrees. Opposite in the sky as in Athens. This day is between Hong Kong's two zenith passages. The altitude is very high, 89.1 degrees.
7. Directly in the center, the same as in Athens.